



IIT ROORKEE



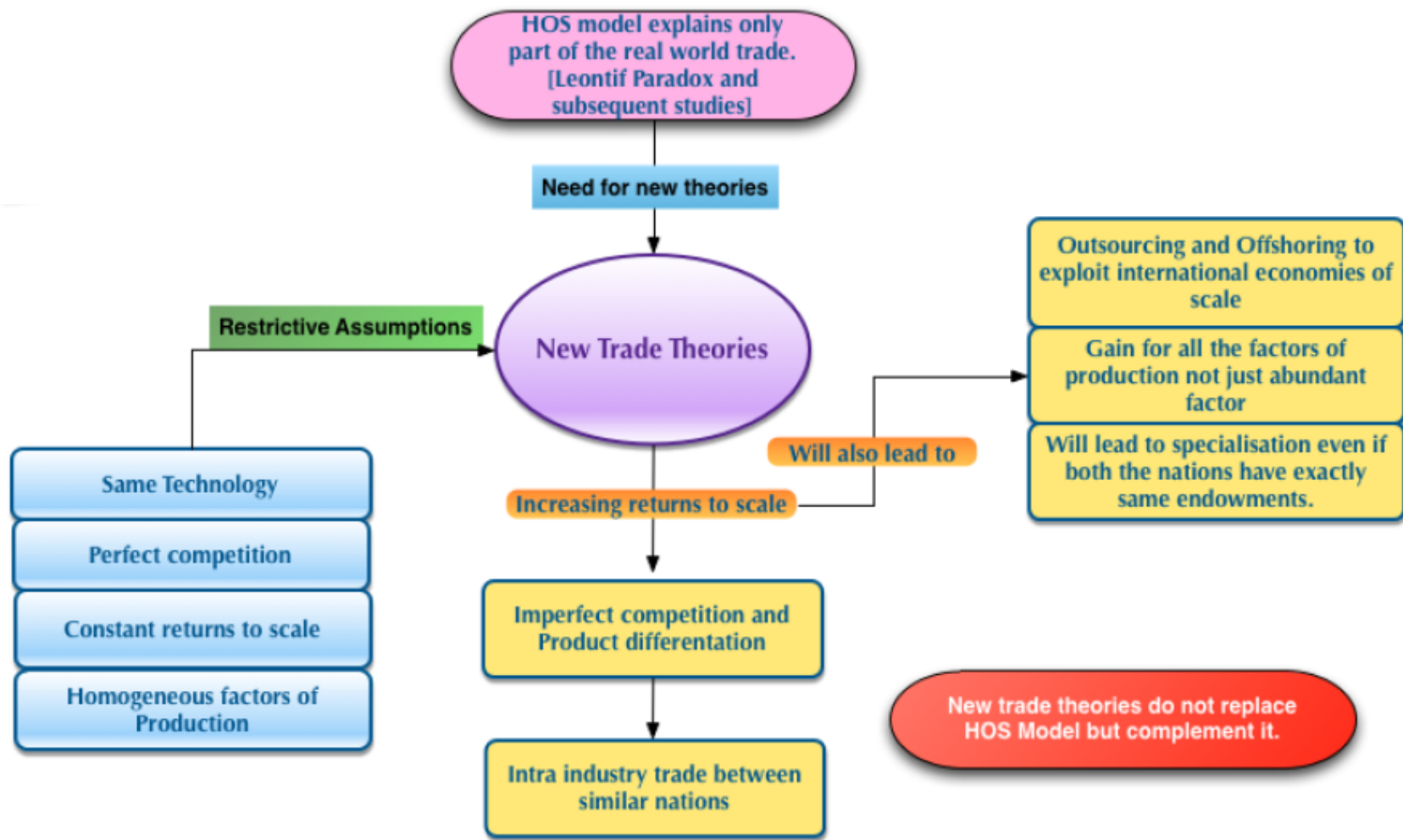
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Intra-industry versus Inter-Industry Trade

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Prologue

- Recent empirical evidences across the globe reveal that the diversification and composition of the export basket of a country are important **determinants of its economic growth** (Aditya and Acharyya 2013; Hausmann et al 2007).
- Empirical evidences **since the 1970s** reveal that a large part of world trade takes place between **similar countries and products** (Grubel and Lloyd 1975), which cannot be explained by the inter-industry trade theories.
- The IIT in a **horizontally differentiated product** can take place when **consumers' utility increases** with variety as in Krugman's (1979) love of variety approach or when consumers view goods as a bundle of characteristics as in the approach of Helpman (1981).
- The quality variation of an export good, means substantive improvement in terms of product features and characteristics – **vertically differentiate products**.
- A higher-end processor-enabled **personal computer** may be an example of such a variation in quality



Justification for IIT

- The reasons for quality differentiation are **income disparity across** and within countries, technological disadvantage and difference in endowment of skilled labour and human capital.
- Quality differences between varieties from the **North and the South** originate from differences in technology (Flam and Helpman (1987) model)
- Generally, it is believed that the **developing countries** are producers and exporters of **poor quality products** (Acharyya 2005; Marjit and Raychaudhuri 1997).
- The conventional belief was that IIT occurs mainly among the **developed countries**.
- Reinert (2003) points out that this phenomenon is emerging even in the developing Asian countries.

- Along with the scarcity of physical and human capital, backward technology and lower rate of innovation, the problems of asymmetric information, moral hazard and adverse selection contribute to the low quality of developing country products.
- The situation has become even worse with **non-tariff barriers** in the form of quality regulations and environmental standards imposed by the developed countries on the imports from the developing countries
- Hypothesis of Hausmann et al (2007) is that “**countries become what they produce.**”
- **Immiserizing growth** is a theoretical situation first proposed by Jagdish Bhagwati, in 1958
- Thus, diversification of exports in high value-added products is required to achieve a more sustained growth effect of openness (Aditya and Acharyya 2013).
- the case with China and India



Types of Economies of Scale

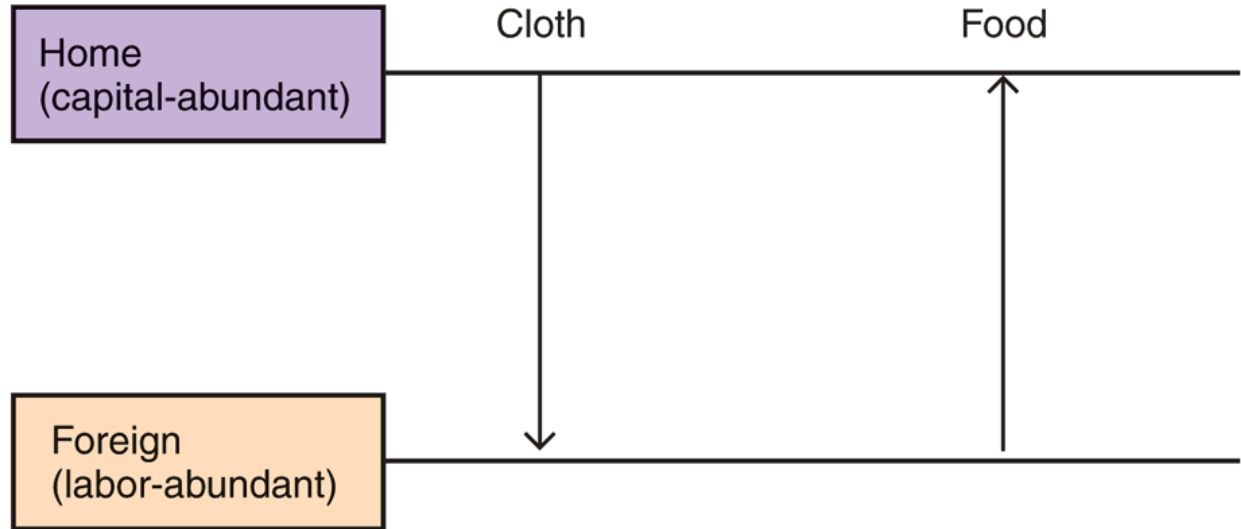
- Economies of scale could mean either that larger firms or that a larger industry (e.g., one made of more firms) is more efficient.
- **External economies of scale** occur when cost per unit of output depends on the *size of the industry*.
- **Internal economies of scale** occur when the cost per unit of output depends on the *size of a firm*.
- External economies of scale may result if a larger industry allows for more efficient provision of services or equipment to firms in the industry.
 - Many small firms that are competitive may comprise a large industry and benefit from services or equipment efficiently provided to the large group of firms.
- Internal economies of scale result when large firms have a cost advantage over small firms, which leads to an imperfectly competitive market.

Inter-industry Trade

Figure 6-6

**Trade in a World
Without Increasing
Returns**

In a world without economies of scale, there would be a simple exchange of cloth for food.



Intra-industry Trade

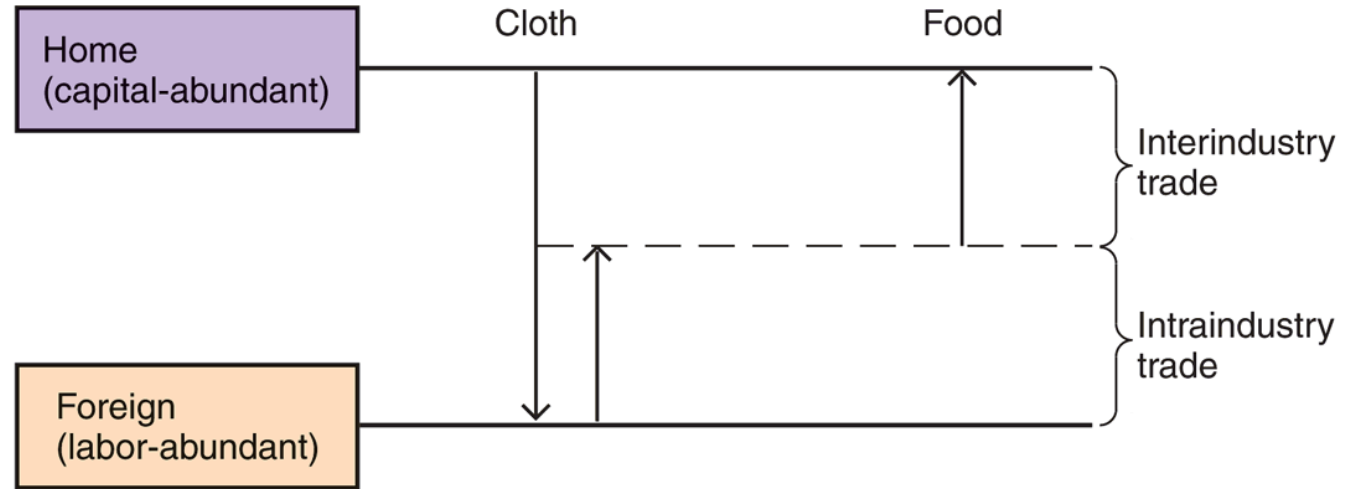


Figure 6-7

Trade with Increasing Returns and Monopolistic Competition

If cloth is a monopolistically competitive industry, Home and Foreign will produce differentiated products. As a result, even if Home is a net exporter of cloth goods, it will import as well as export cloth, giving rise to intraindustry trade.

Intra- and Inter-industry Trade

- **Inter-industry trade:** countries export goods of one product category and imports goods of other product category as in the **Ricardian** and the **Heckscher-Ohlin** models
 - Basis for trade: **Comparative Advantage** due to differences in productivity (Ricardian model) or in factor endowments (HO-model)
- **Intra-industry trade:** countries export and import products of the same products category as in the **Krugman model**
 - Basis for trade: **Internal economies of scale**

IIT and India

- Even though India is an unskilled-labour abundant country, its export are largely to the skilled-labour abundant countries (Acharyya 2013).
- China and India are identified as outliers among developing countries (Rodrik 2006) in terms of the productivity of the export basket.
- Rodrik (2006) expected India's EXPY (which measures the productivity of export basket) to be even higher with software exports.
- Bagchi et al (2014) for six Indian industries over 1990–2013 and Srivastava and Medury (2011) during 2000–08.
- A significant proportion of IIT now takes place in intermediate and final goods. (example of motor parts and passenger cars)
- The share of intermediate goods dominate IIT globally (Brulhart 2008; Marini 2017) among other categories.

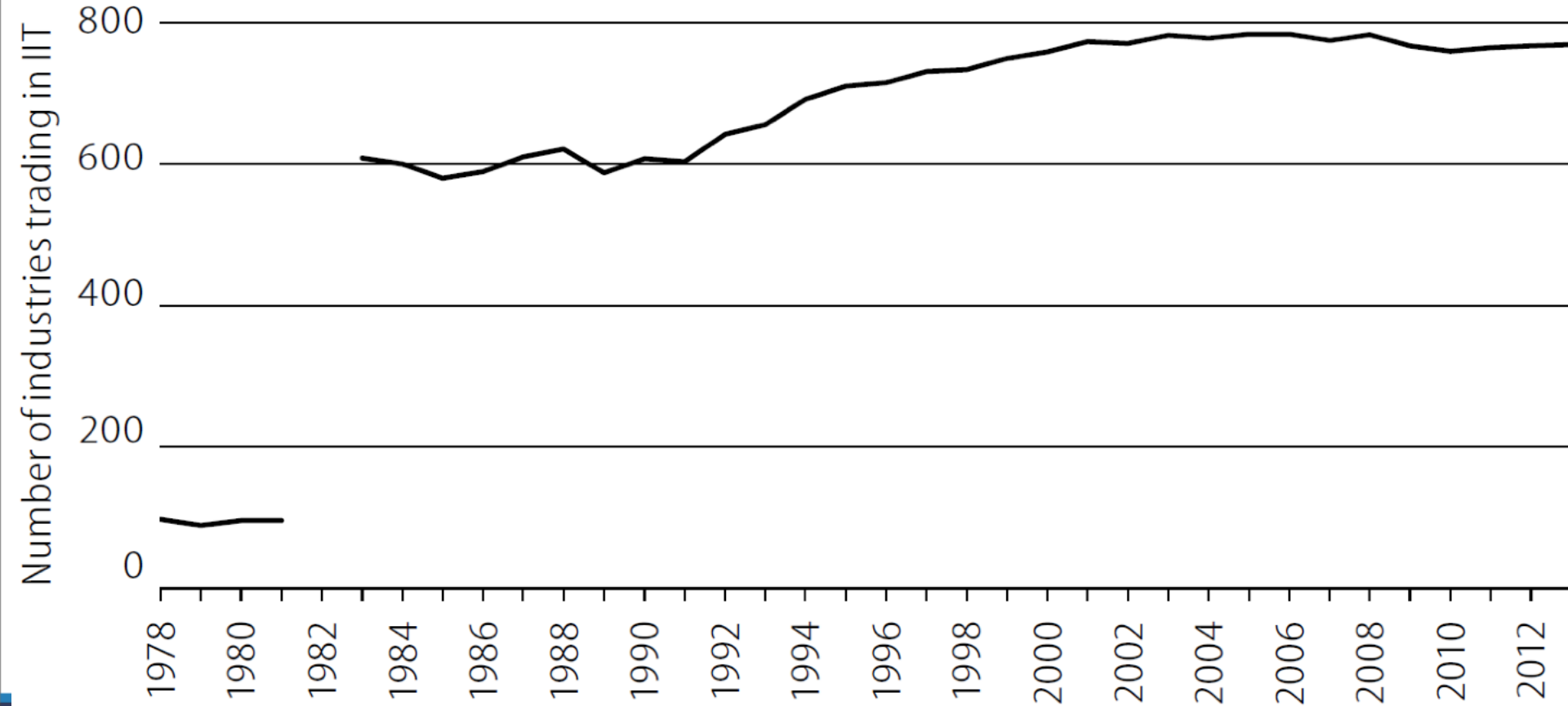
Types of IIT

- Trade in varieties of a product characterised by different attributes is HIIT (Horizontal) and when it is characterised by quality difference, such trade is VIIT (Vertical).
- Recent studies reveal that there has been a steady increase in India's IIT, especially in the post-reform era (Veeramani 2002, 2004; Burange and Chaddha 2008).
- In particular, whether it is the horizontal differentiation or vertical differentiation that dominates India's IIT basket
- Interesting to know, which sectors has there been a transition from HIIT to VIIT or vice versa.
- The horizontal models are considered to be more relevant for understanding the occurrence of IIT among the developed countries and the vertical models for explaining IIT between unequal partners.

Approaches to Measure

- Bagchi et al (2014) used the **unit value dispersion criterion** following Greenaway, Hine and Milner (**GHM**) approach at **HS-6-digit** level
- This GHM approach disentangles total IIT into horizontal and vertical,
- termed as, high vertical or *h-VIIT* and inferior, as, low vertical or *l-VIIT* products
- Bagchi et al (2014) estimated IIT for six major Indian manufacturing industries over the period 1990–2013.
- They found that technologically inferior quality products (or *l-VIIT*) have been dominant in India's export basket throughout the period of study,
- whereas HIIT and export of high technology goods (*h-VIIT*) have gained some momentum after the global economic slowdown in 2008

IIT in India



Data: World Bank's World Integrated Trade Solution (WITS)

*Thank
you*

